Experiment 34: KNN Classification for Medical Treatment

Aim: To predict medical treatment outcomes using KNN and evaluate model performance.

Algorithm:

1. Import KNN and evaluation metrics.
2. Create patient data.
3. Split data into training and testing sets.
4. Train the KNN classifier.
5. Predict treatment outcomes.
6. Calculate accuracy, precision, recall and F1-score.

Code:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

X = [[20, 120], [25, 130], [30, 140], [35, 150],
     [45, 160], [50, 170], [55, 180], [60, 190]]

y = [0, 0, 0, 0, 1, 1, 1, 1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("Predictions:", pred)
print("Accuracy:", accuracy_score(y_test, pred))
```

print("Precision:", precision_score(y_test, pred, zero_division=0))

print("Recall:", recall_score(y_test, pred, zero_division=0))

print("F1 Score:", f1_score(y_test, pred, zero_division=0))

Input:

Patient features: Age and Blood Pressure

Output:

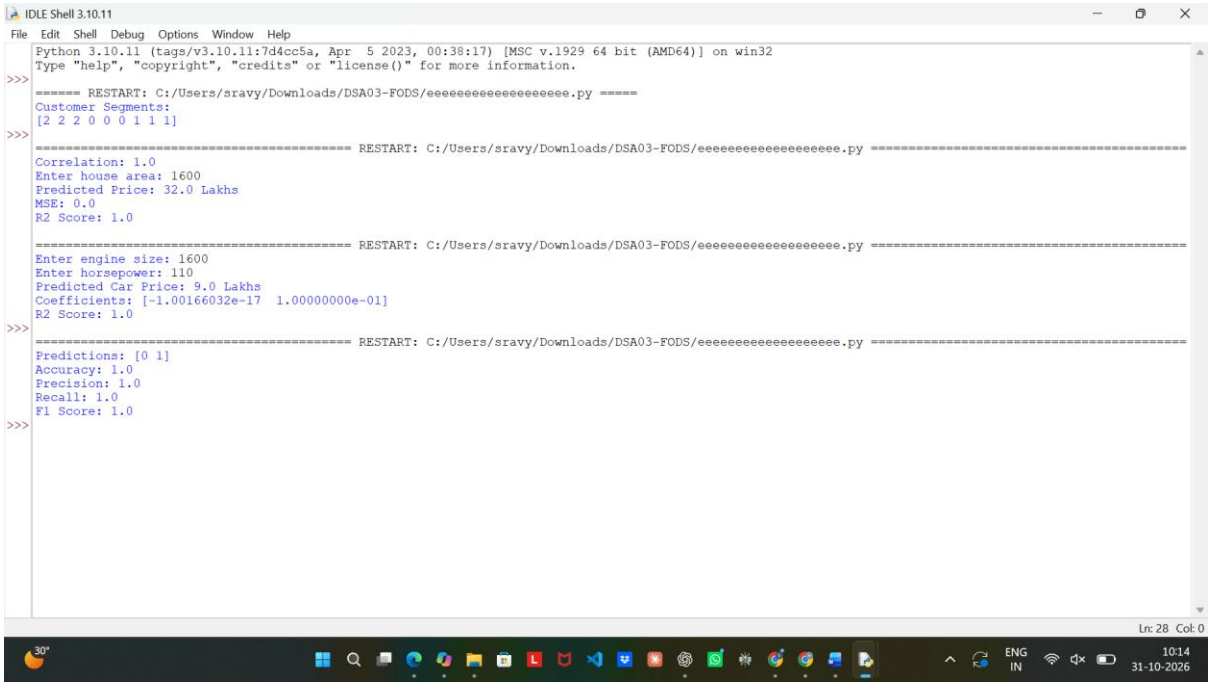
Predictions: [0 1]

Accuracy: 1.0

Precision: 1.0

Recall: 1.0

F1 Score: 1.0



Result: Treatment outcomes were predicted and evaluated using KNN.